A.7 Chapter 7: Maximizing throughput with batch optimization

There are two exercises in this chapter:

1. The first covers the implementation of Thompson sampling under the batch setting.
2. The second shows us how to run Bayesian optimization policies on a four-dimensional aero-structural optimization problem.

A.7.1 Extending Thompson sampling to the batch setting via resampling

Remember that TS in the sequential setting, which we learn in Chapter 5's Section 5.3, draws one sample from the current Gaussian process (GP) belief about the objective function and queries the data point that maximizes that sample. In the batch setting, we simply repeat this process of sampling from the GP and maximizing the sample multiple times to assemble a batch of queries of the desired size. The code for this exercise can be found in `CH07/02 - Exercise 1.ipynb`.

1. Recreate the batch Bayesian optimization loop in `CH07/01 - Improvement and confidence bound policies.ipynb`.
2. Implement Thompson sampling with a Sobol sampler as described in Section 5.3 of Chapter 5.
   We implement the policy as follows, where we use a 2000-element Sobol sequence and specify the number of samples as the batch size:

```
num_candidates = 2000 ❶

... ❷

for i in tqdm(range(num_iters)):
    ... ❸

    sobol = torch.quasirandom.SobolEngine(2, scramble=True) ❹
    candidate_x = sobol.draw(num_candidates) ❹
    candidate_x = (bounds[1] - bounds[0]) * candidate_x + bounds[0] ❹

    ts = botorch.generation.MaxPosteriorSampling(model, replacement=False)
    next_x = ts(candidate_x, num_samples=batch_size) ❺

    ... ❻
```

   ❶ Specify the length of the Sobol sequence.

   ❷ Randomly pick the initial training data.

   ❸ Retrain the Gaussian process.

   ❹ Initialize the Sobol sequence.

   ❺ Draw multiple samples from the Gaussian process.

   ❻ Query the objective function and update the training data.

3. Run this Thompson sampling policy on the hyperparameter tuning objective function and observe its performance.

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